

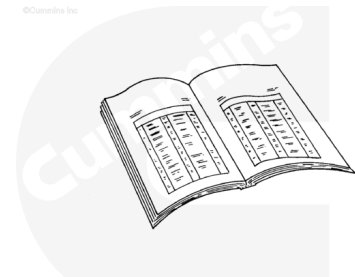
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

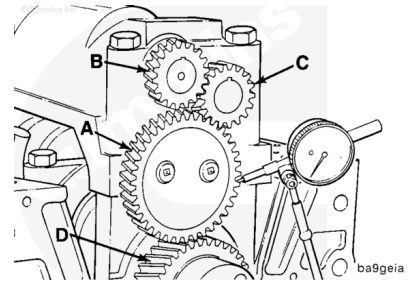
- Disconnect the batteries. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/40/40-013-009.html>).
- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>).
- Remove the lubricating oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>).
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>).

Remove

Measure Backlash

Use an indicator to measure the backlash of the idler gear (A), upper shaft gear (B), and lower shaft gear (C).

Note : Do not allow the adjoining gear to move or the reading will be the total of both gears.

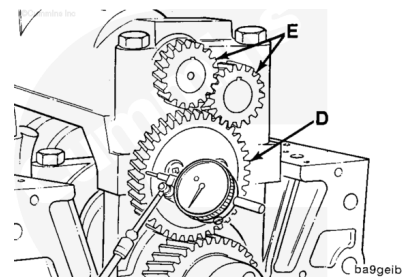


	mm		in
Idler (A) to (D)	0.088	MIN	0.003
	0.420	MAX	0.017
Upper Shaft (B) to (A)	0.153	MIN	0.006
	0.355	MAX	0.014
Lower Shaft (C) to (B)	0.088	MIN	0.003
	0.420	MAX	0.017

Record for use during inspection.

Measure End Play

Use an indicator to measure the end play of the idler gear (D) and shaft (E).

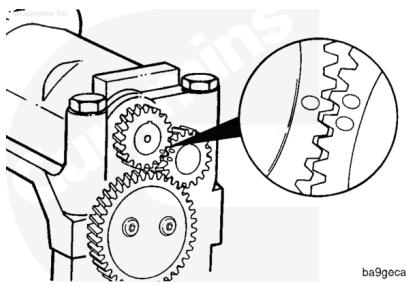


	mm		in
(D)	0.130	MIN	0.005
	0.630	MAX	0.024
(E)	0.075	MIN	0.003
	0.175	MAX	0.007

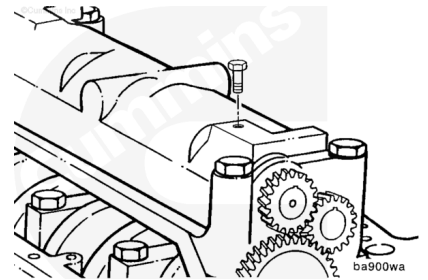
Record for use during inspection.

Lock Balancer

Rotate the balance gears until the timing marks are aligned.



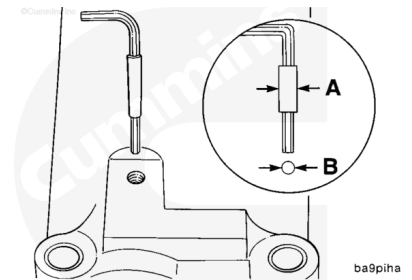
If the balancer shaft has a tapped hole, the shaft can be locked in position by temporarily installing a M8 capscrew through the housing and into the shaft.



Follow this procedure if the shaft does **not** have a tapped hole.
Wrap the 4.5-mm Allen wrench with 1-inch wide masking tape until it has a snug fit in the hole in the balancer housing.

A = Approximately 10 mm [0.4 in]

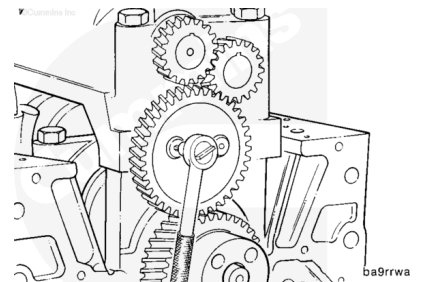
B = 10 mm [0.4 in]



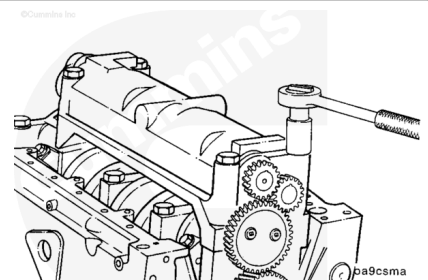
Remove Balancer

Loosen the socket head capscrew for the balancer idler gear retainer using an 8-mm Allen wrench.

Do **not** remove the capscrews.

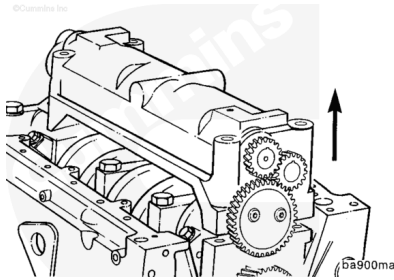


Remove the number 1 and number 4 main bearing capscrews.



Move the idler retainer until the pin in the number 1 cap disengages the slot in the retainer.

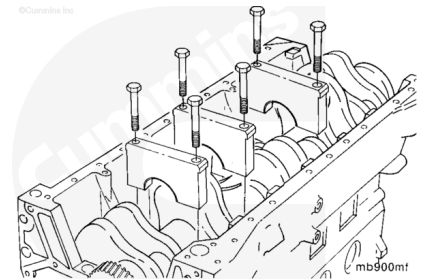
Remove the balancer assembly.



Install

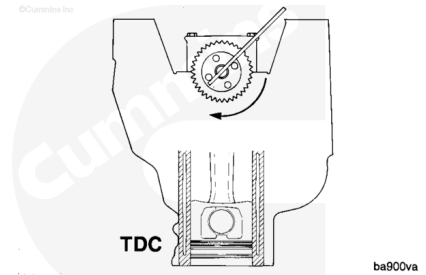
Use a 23-mm wrench to rotate the cylinder block on the relaxed stand until the crankshaft is positioned at the top and parallel to the floor.

The number 1 and number 4 main bearing capscrews **must** be removed to install the balancer.



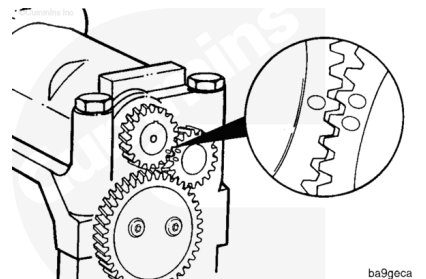
Rotate the crankshaft until the number 1 piston is at top dead center.

The engine **must** have a cylinder at top dead center for correct gear teeth alignment when the balancer is installed.



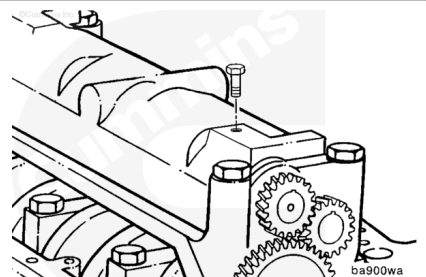
Rotate the balancer gears until the timing marks are aligned.

The balancer **must** be kept in this position for correct installation on the engine.



If the balancer shaft has a tapped hole, the shaft can be locked into position by temporarily installing an M8 capscrew through the housing and into the shaft.

Use a 13 mm wrench.



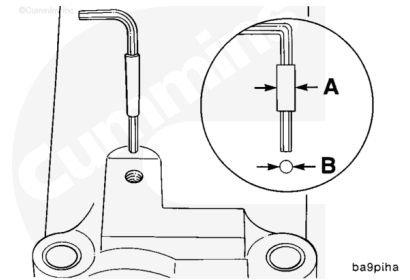
The idler gear retainer capscrews **must** be loose.

Follow this procedure if the shaft does **not** have a tapped hole.

Wrap the 4.5-mm Allen wrench with 1-inch wide masking tape until it has a snug fit in the hole in the balancer housing.

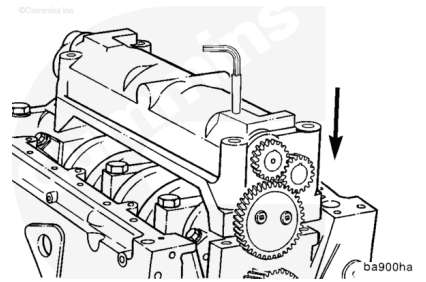
A = Approximately 10 mm [0.4 in]

B = 10 mm [0.4 in]



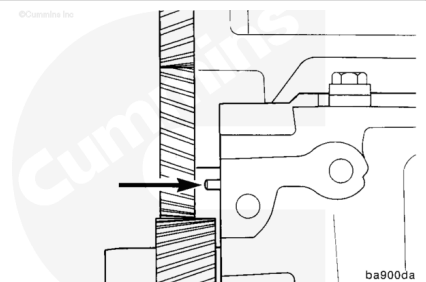
Position the locked balancer assembly onto the main bearing caps.

The assembly **must** be located squarely with the alignment ears against the side of the caps.

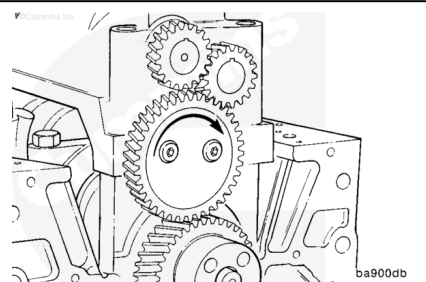


Align the slot in the idler gear retainer with the locating pin in the main bearing cap.

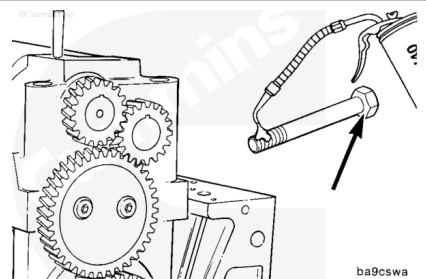
Slide the balancer into position.



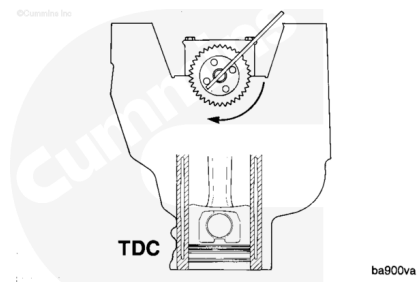
The idler gear can be slightly rotated to aid in alignment.



Lubricate the main bearing capscrew threads and the underside of the capscrew heads with clean engine oil.

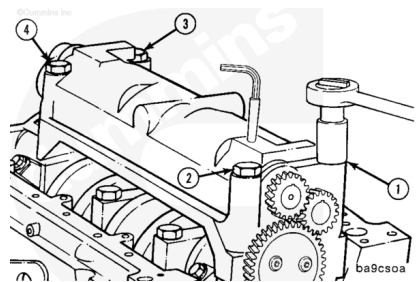


If the capscrews do **not** install freely, check to be sure the engine has a piston at top dead center.

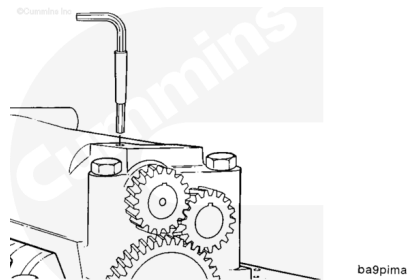


Tighten the capscrews evenly, use a 23 mm wrench, follow the illustrated sequence.

Torque Value:	Step 1	60 n.m [44 ft-lb]
	Step 2	119 n.m [88 ft-lb]
	Step 3	176 n.m [130 ft-lb]

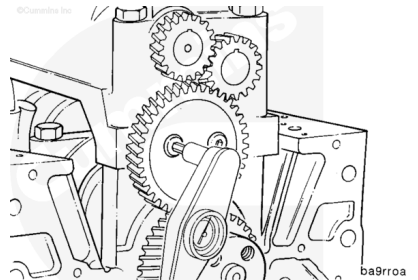


Remove the locking capscrew or Allen wrench from the balancer.



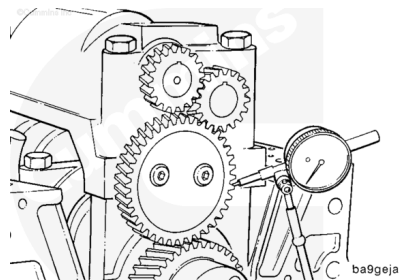
Tighten the idler gear retainer capscrews, use an 8-mm Allen wrench.

Torque Value: 57 n•m [42 ft-lb]



Measure the idler gear backlash.

mm		in
0.088	MIN	0.003
0.420	MAX	0.017

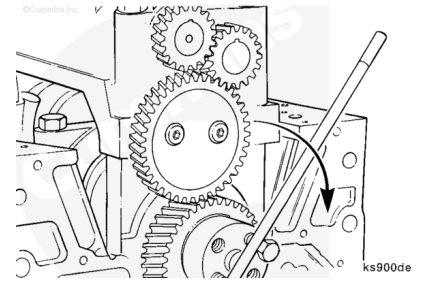


If the idler gear does **not** meet the specifications, loosen the idler gear retainer capscrews.

Position the idler gear and tighten the capscrews.

Torque Value: 57 n•m [42 ft-lb]

If the crankshaft does **not** rotate freely, make sure the balancer does **not** have an interference.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the lubricating oil suction tube. Refer to Procedure 007-035 (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>).
- Install the lubricating oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>).
- Fill the engine with lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>).
- Connect the batteries. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/40/40-013-009.html>).